

ADVANCED DATA STRUCTURES

Common to CSE&IT

III Semester

Course Code:201CS3T01

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Course Outcomes:**At the end of the Course, Student will be able to:**

- CO1:** Make use of external sorting and hashing for sorting and searching.
CO2: Illustrate the operations of Priority Queues.
CO3: Analyze the operations of efficient Binary Search Trees and multiway search trees.
CO4: Compare different operations of digital search structures.
CO5: Apply the string processing algorithms for pattern matching.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	1
CO2	2	3	-	-	-	-	-	-	-	-	-	-
CO3	2	3	-	-	-	-	-	-	-	-	-	-
CO4	2	3	-	-	-	-	-	-	-	-	-	-
CO5	3	2	-	-	-	-	-	-	-	-	-	1

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2
CO1	2	-
CO2	1	-
CO3	2	-
CO4	1	-
CO5	2	-

Unit - I

External Sorting: Introduction, K-way Merge Sort, Buffer Handling for parallel Operation, Run Generation, Optimal Merging of Runs, Huffman Tree.

Hashing: Introduction to Static Hashing, Hash Tables, Hash Functions, Different Hash Functions, Collision Resolution Techniques, Dynamic Hashing.

Unit – II

Priority Queues (Heaps): Introduction, Binary Heaps-Model and Simple Implementation, Basic Heap Operations, Other Heap Operations, Applications of Priority Queues, Binomial Heaps (or Queues), Binomial Heap Structure and Implementation, Binomial Queue Operations.

Unit – III

Efficient Binary Search Trees: Self-balancing Binary Search Tree, AVL Trees, Rotations-LL, RR, LR and RL, Searching, Insertion, Deletion operations on AVL Trees, Red-Black Tree, Properties and Representation of Red-Black Trees, Insertion and deletion operations on Red-Black Trees, Applications of Red-Black Trees

Unit – IV

Multiway Search Trees: M-Way Search Trees Definition and Properties, B-Tree Definition and Properties, Searching, Insertion and Deletion operations on B-Trees, B+ Tree, Insertion and Deletion operations on B+ Trees.

Digital Search Structures: Introduction to Digital Search Tree, Operations on Digital Search Trees- Insertion, Searching, and Deletion.

Unit – V

Digital Search Structures: Binary Tries, Compressed Binary Trie, Patricia, Searching Patricia, inserting into Patricia, delete a node from Patricia, Multiway Tries- Definition, Searching a Trie, Compressed Tries, Compressed Tries with Digit Numbers-Searching, Insertion, Deletion.

String Processing: String Operations, Brute-Force Pattern Matching, The BoyerMoore Algorithm, The Knuth-Morris-Pratt Algorithm, The Longest Common Subsequence Problem (LCS).

Text Books:

1. Advanced Data Structures, ReemaThareja, S. Rama Sree, Oxford University Press.
2. Data Structures and Algorithm Analysis in C, Mark Allen Weiss, Second Edition, Pearson.

Reference Books:

1. Fundamentals Of Data Structures In C, Horowitz, Sahni, Anderson-Freed, Second edition
2. Data Structures and Algorithms, A. V. Aho, J. E. Hopcroft, and J. D. Ullman, Pearson.
3. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson and Ronald L. Rivest, Third Edition, The MIT Press.
4. Advanced Data Structures, Peter Brass, Cambridge University Press.

Web Links:

1. https://ocw.mit.edu/courses/...and...data-structures...notes/MIT6_851S12_L1.
2. <http://nptel.ac.in/courses/106103069/26>
3. <https://csd.cs.cmu.edu/course-profiles/15-121-Introduction-to-Data-Structures>
4. <https://www.cs.purdue.edu/cgvlab/courses/251/lectures/slides/04.03-PatternatchingAndTries.pdf>
5. <https://www.csie.ntu.edu.tw/~ds/ppt/ch5/chapter5.PPT>